KEY TOPIC 4 • PUPIL WORKBOOK

Medicine in Modern Britain

c1900–present: Ideas on Causes, Diagnostic Technology, The NHS, Penicillin & Lung Cancer



Primary Visual Evidence: *Health Minister Aneurin Bevan visiting Sylvia Beckingham (first NHS patient), Park Hospital, Davyhulme, 5 July 1948* CENTRAL OFFICE OF INFORMATION ARCHIVE

PUPIL NAME: _____ TARGET: _____ CLASS: _____

KEY TOPIC 4 SPECIFICATION FRAMEWORK (1HI0/11)	LESSON DONE	QUIZ [10]
Key Topic 4.1: Ideas on Causes: Genetics, DNA & The Human Genome Project *How did Rosalind Franklin's Photograph 51, Watson & Crick, and genomic sequencing unlock hereditary disease?*	☐	☐
Key Topic 4.2: Lifestyle Factors & The Technological Revolution in Diagnosis *How did the shift to chronic lifestyle illnesses, Röntgen's X-rays, and Hounsfield's CT scanner eliminate exploratory surgery?*	☐	☐
Key Topic 4.3: Magic Bullets, High-Tech Treatments & The Birth of the NHS *How did Salvarsan 606, Prontosil, and Aneurin Bevan's 1948 NHS create free healthcare for every British citizen?*	☐	☐
Key Topic 4.4: Case Study 1: The Antibiotic Revolution: Fleming, Florey & Chain *How did Fleming's 1928 discovery, the Oxford team's purification, and American deep-tank vats deliver 2.3m doses for D-Day?*	☐	☐
Key Topic 4.5: Case Study 2: Public Health & The Fight Against Lung Cancer *How did Doll & Hill prove the tobacco link, and how did ad bans, the 2007 public smoking ban, and plain packaging end laissez-faire?*	☐	☐

Living Timeline • Part 1: Magic Bullets, Blood Groups & Early Antibiotics (1900–1940)

Living Timeline Dual-Coding Protocol: Draw visual symbols for ABO agglutination, Salvarsan magic bullets, Fleming's mould petri dish, and Prontosil sulfa dye across the timeline below.

1901 • Karl Landsteiner: ABO Blood Groups Discovered

Key Topic 4.1

Landsteiner identifies A, B, and O blood groups, proving agglutination occurs when incompatible blood types mix. This discovery transforms blood transfusions from fatal gambles into safe, predictable surgical lifelines, foundational for modern trauma surgery.

1909 • Paul Ehrlich & Sahachiro Hata: Salvarsan 606

Key Topic 4.3

Building on dye research, Ehrlich and Hata systematically test hundreds of arsenic compounds, discovering arsphenamine (compound 606). Humanity's first synthetic chemical 'magic bullet' targets syphilis bacteria without poisoning human tissue.

1928 • Alexander Fleming: Accidental Discovery of Penicillin

Key Topic 4.4

Returning from holiday, Fleming notices a contaminated staphylococcus culture dish with a halo of destroyed bacteria around Penicillium notatum mould. Fleming publishes in 1929, though unable to purify or extract enough active substance for clinical trials.

1932 • Gerhard Domagk: Prontosil & Sulfonamide Drugs

Key Topic 4.3

Domagk demonstrates that the red leather dye Prontosil cures streptococcal blood poisoning in mice (and saves his own daughter's arm). Chemists isolate the active ingredient sulfonamide, launching the world's first mass-prescribed class of antibacterial wonder drugs.

Living Timeline Dual-Coding Protocol: Draw visual symbols for Peoria deep fermentation vats, the 1948 NHS charter, Franklin's Photograph 51 double helix, and CT scan slices across the timeline below.

1941–44 • Florey, Chain & Heatley: Penicillin Mass Production

Key Topic 4.4

Howard Florey and Ernst Chain purify penicillin at Oxford, successfully treating mice and Albert Alexander (1941). Heatley and Florey travel to Peoria, Illinois, using corn-steep liquor and deep-tank fermentation to produce 2.3 million doses for D-Day casualties.

1948 • Aneurin Bevan: Foundation of the NHS

Key Topic 4.3

On 5 July 1948, the National Health Service opens at Park Hospital, Manchester. Championed by Bevan and the 1942 Beveridge Report, it nationalises voluntary and municipal hospitals, providing universal healthcare free at the point of clinical delivery.

1953 • Watson, Crick & Franklin: DNA Double Helix Structure

Key Topic 4.1

Using Rosalind Franklin and Maurice Wilkins's X-ray crystallography (Photograph 51), Watson and Crick deduce the double-helix structure of DNA at Cambridge, unlocking hereditary genetic code and modern molecular gene therapy.

1971–2003 • CT Scanners (1971) & The Human Genome Project (2003)

Key Topic 4.2 • 4.1

Godfrey Hounsfield invents the EMI CT scanner, combining computer algorithms with X-rays to image internal organs without surgery. In 2003, the international Human Genome Project maps all 3 billion chemical base pairs in human DNA, founding precision medicine.

Why did the discovery of DNA revolutionize understanding of disease causes, yet take decades to yield treatments?

Core Specification Focus: Genetics and DNA: Mendel; Rosalind Franklin (Photograph 51), Watson and Crick (double helix 1953); the Human Genome Project (1990–2003); genetic screening and hereditary disease.

1 c1900–1940s • Why did doctors assume proteins carried hereditary traits? [1.1]

• Growing wrinkled green garden peas	
• Spotting dark rod shapes inside cells	
• Assuming heavy protein chains hold secrets	

2 1951–1952 • How did Photograph 51 prove DNA's helical structure? [1.2]

• Beaming invisible rays through wet DNA fibers	
• Exposing glass photographic plates for sixty hours	
• Revealing clear fuzzy X-shaped shadow marks	

3 1953 • How did base pairing explain hereditary replication? [1.3]

• Twisting thin metal wire into spirals	
• Locking four chemical pairs like puzzle pieces	
• Rushing to celebratory pints at The Eagle pub	

4 1990–2003 • Why was mapping 3 billion base pairs a medical milestone? [1.4]

• Linking laboratory supercomputers across six nations	
• Reading three billion chemical letters in blood	
• Publishing complete human code free online	

5 2000s–Present • Why did understanding causes outpace finding cures? [1.5]

• Testing saliva drops for faulty breast genes	
• Matching targeted pills to personal body code	
• Struggling to fix broken genetic code inside bodies	

Franklin's 62-hour X-ray exposure revealed the double helix; Watson and Crick celebrated in The Eagle pub.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why there was rapid progress in understanding the causes of disease in the period c1950 to the present. [12 marks]

Stimulus: **Rosalind Franklin's Photograph 51 (1952)** | **The Human Genome Project (1990–2003)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Gregor Mendel • Rosalind Franklin • Photograph 51 (1952) • Watson & Crick • double helix model • Human Genome Project (1990–2003) • gene screening • hereditary disease

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.1). In the drawing box, sketch Franklin's Photograph 51 cross pattern and Watson & Crick's double helix model. Annotate how base pairing unlocked the causes of hereditary disease! **[ ] Done**

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which scientific methods for investigating disease causes in the late twentieth century (c1950–c2000) were different from methods used in the late nineteenth century (c1870–c1900). [4 marks]

Contrast computerized DNA sequencing with manual microscopic staining with chemical dyes.

Were high-tech diagnostic scanners more vital to modern health than state lifestyle campaigns?

Core Specification Focus: Lifestyle factors and diagnosis: shift from infectious to chronic disease; non-invasive diagnosis: Röntgen (X-rays), CT scanners (Hounsfield 1971), MRI; endoscopes, blood and ultrasound testing; lifestyle campaigns.

1 1895–1918 • How did X-rays transform surgical trauma diagnosis in WWI? [2.1]

- **Photographing wife Bertha's hand and gold wedding ring**
- **Driving mobile medical vans to muddy trenches**
- **Finding jagged lead shrapnel without deep cuts**

2 1930s–1960s • Why was fiber-optic endoscopy superior to exploratory surgery? [2.2]

- **Spinning red test tubes in fast centrifuges**
- **Counting abnormal white cells on glass slides**
- **Sliding flexible glass cables down throat tubes**

3 1971–1980s • Why did CT scans transform internal soft-tissue diagnosis? [2.3]

- **Spinning heavy donut scanners around hospital beds**
- **Combining computer pictures into 3D body slices**
- **Using magnetic radio waves to find brain lumps**

4 c1950–1980s • What drove the shift from infectious to lifestyle diseases? [2.4]

- **Banishing deadly childhood fevers with wonder drugs**
- **People living past seventy years of age**
- **Rising chest tumours and clogged heart arteries**

5 1980s–Present • Why do governments favour lifestyle prevention over treatment? [2.5]

- **Clicking compulsory car seatbelts across chests**
- **Extinguishing glowing cigarettes outside pub doors**
- **Taxing sugary fizzy cans and advertising fruit**

Bertha Röntgen cried "I have seen my death!" upon seeing her bones; doctors saw the dawn of non-invasive diagnosis.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'Technological innovations in medical diagnosis were more important than public health lifestyle campaigns in improving health in Britain c1900–present.' How far do you agree? [16+4 marks]

Stimulus: **Wilhelm Röntgen's X-rays (1895)** | **The 2007 smoking ban in public places** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, high-tech diagnosis was vital because... • For example, ... • This was significant because... • On the other hand, lifestyle campaigns mattered because... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

Wilhelm Röntgen (1895) • X-rays • CT scanner (1971) • Godfrey Hounsfield • MRI • non-invasive imaging • 2007 smoking ban • Change4Life

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.2). In the drawing box, sketch Bertha Röntgen's hand X-ray and Godfrey Hounsfield's circular CT scanner. Annotate how scanning permanently ended blind exploratory surgery! **[ ] Done**

Hounsfield's CT scanner combined rotating X-rays with digital computers to slice through soft body tissue.

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which methods of diagnosing illness in the late twentieth century (c1970–present) were different from methods used in the nineteenth century (c1800–c1900). [4 marks]

Contrast non-invasive 3D internal scanning and blood labs with external observation and exploratory surgery.

Was the establishment of the NHS in 1948 more transformative than the development of magic bullets?

Core Specification Focus: Magic bullets and healthcare: Ehrlich (Salvarsan 606), Domagk (Prontosil); high-tech treatments: dialysis, hip replacements; Beveridge Report (1942); Bevan and the foundation of the NHS (1948); universal healthcare.

1 1906–1909 • Why was Salvarsan 606 termed the first "magic bullet"? [3.1]

- **Testing hundreds of toxic yellow arsenic mixtures**
- **Hunting chemical bullets that only hit germs**
- **Curing deadly human syphilis with magic bullet**

2 1932–1935 • How did Prontosil prove synthetic drugs could kill microbes? [3.2]

- **Injecting bright red leather dye into mice**
- **Saving sick daughter from arm-amputation death**
- **Discovering active white sulfa medicine powder**

3 1942 • How did WWII unity create momentum for state healthcare? [3.3]

- **Listing Five Giants that crush poor workers**
- **Surviving Nazi air bombs with community spirit**
- **Demanding free medical care for every citizen**

4 1945–1948 • How did Bevan compromise to persuade resistant doctors? [3.4]

- **Offering private bed perks to stubborn doctors**
- **Delivering free spectacles and dentures to families**
- **Abolishing terrifying doctor bills forever**

5 1950s–Present • Why has modern medical success created financial strain? [3.5]

- **Pumping blood through giant artificial kidney filters**
- **Replacing worn hip joints with metal balls**
- **Charging small fees for paper medicine slips**

Ehrlich tested 605 failed arsenic compounds before compound 606 finally cured syphilis in 1909.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'The establishment of the National Health Service in 1948 was the most significant turning point in medical care in the period c1900–present.' How far do you agree? [16+4 marks]

Stimulus: **The National Health Service (1948)** | **The discovery of Salvarsan 606 (1909)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, the NHS was transformative because... • For example, ... • This was significant because... • On the other hand, magic bullets were crucial because... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

Paul Ehrlich • Salvarsan 606 (1909) • Gerhard Domagk • Prontosil (1932) • William Beveridge (1942) • Aneurin Bevan • NHS (1948) • free at point of delivery

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.3). In the drawing box, sketch Ehrlich's test tube of Salvarsan 606 and the 1948 NHS information leaflet. Annotate how free healthcare abolished medical poverty! **[ ] Done**

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE DIFFERENCE [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which healthcare access in Britain after 1948 was different from healthcare access in the 1930s. [4 marks]

Contrast free universal healthcare funded by taxation with private fees and voluntary charity hospitals.

Why did penicillin require a global war and American industrial might to become a mass-produced cure?

Core Specification Focus: Case study: The Antibiotic Revolution: Fleming's discovery (1928); Florey and Chain (Oxford purification 1938–40); wartime need and US mass production (Peoria deep-tank fermentation 1941–44); D-Day supply; antibiotic resistance.

1 1928–1929 • Why could Fleming not turn mould into a clinical medicine? [4.1]

• Finding fluffy mould on dirty glass dish	
• Golden germs dissolving in clear circle halos	
• Failing to separate fragile liquid juice	

2 1938–1940 • How did Norman Heatley solve the purification challenge? [4.2]

• Brewing fungus inside hospital bedpans and milk churns	
• Winning American dollar grants during air raids	
• Freezing brown liquid into precious yellow crystals	

3 1940–1941 • Why was Alexander's trial both a success and a tragedy? [4.3]

• Infecting eight mice and keeping treated four alive	
• Scraping rose thorn scratches on policeman's face	
• Running out of drops before patient recovers	

4 1941–1944 • How did US industrial fermentation enable mass production? [4.4]

• Feeding corn beer wash to rotting melon mould	
• Pumping sterile air through giant ten-thousand-gallon vats	
• Brewing million doses for D-Day landing craft	

5 1944–Present • Why did Fleming warn that antibiotic overuse causes resistance? [4.5]

• Spraying white powder directly into torn soldiers' limbs	
• Three scientists sharing golden Nobel prize medal	
• Overusing capsules creating tough hospital superbugs	

Fleming almost washed his petri dish in Lysol, but noticed a clear halo where mould destroyed golden staphylococci.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why penicillin was successfully developed and mass-produced in the period 1928–1945. [12 marks]

Stimulus: **Alexander Fleming's discovery (1928)** | **The impact of the Second World War** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...

KEY VOCABULARY BANK:

Alexander Fleming (1928) • penicillin • Howard Florey • Ernst Chain • Norman Heatley • Albert Alexander (1941) • Peoria fermentation • mass production (1944)

Write your response in the space provided below:

(Response continues on facing page)

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.4). In the drawing box, sketch Fleming's contaminated petri dish and an American 10,000-gallon deep-tank vat. Annotate how penicillin saved Allied troops on D-Day! **[ ] Done**

The Oxford team grew penicillin in bedpans; American factories brewed it in 10,000-gallon fermentation vats.

Continue your response to Question 4 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** SCORE: [/ 4] • 5 MINS

Explain one way in which Alexander Fleming's discovery of penicillin in 1928 was similar to Edward Jenner's discovery of the smallpox vaccine in 1796. [4 marks]

Focus on both breakthroughs relying on accidental observations of natural organisms protecting against disease.

Why has the modern fight against lung cancer required state coercion rather than laboratory cures?

Core Specification Focus: Case study: Fight against lung cancer: epidemiology of Doll and Hill (1950); diagnosis: chest X-rays, CT, bronchoscopy; treatments: surgery, radiotherapy, chemotherapy, immunotherapy; government prevention: ad bans, 2007 smoking ban, plain packs (2016).

1 1920s–1950 • Why did early doctors fail to link smoking to lung cancer? [5.1]

- **Men coughing black blood into hospital basins**
- **Blaming smelly road tarmac and lorry exhaust fumes**
- **Colourful billboards showing smiling smoking doctors**

2 1950–1954 • How did statistical epidemiology prove the causal link? [5.2]

- **Interviewing sick patients across twenty London wards**
- **Tracking forty thousand real family doctors over years**
- **Proving heavy smokers face fifty-times higher risk**

3 1962–1971 • Why did government action encounter strong tobacco lobbying? [5.3]

- **Banning glowing cigarette commercials on evening television**
- **Printing bold black warning text on cardboard cartons**
- **Sponsoring fast racing cars to keep brands famous**

4 2000s • Why was passive smoking crucial in banning public smoking? [5.4]

- **Breathing toxic secondhand smoke in crowded rooms**
- **Padlocking pub ashtrays and stepping outside into cold**
- **Raising legal counter buying age to eighteen**

5 2010s–Present • Why does low cancer survival prove prevention beats cure? [5.5]

- **Selling plain ugly mud-green packets with graphic photos**
- **Guiding tiny robotic camera pincers between chest ribs**
- **Boosting immune soldier cells to attack hidden tumours**

Doll and Hill proved smoking caused lung cancer in 1950; Doll immediately extinguished his pipe and lived to age 92.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

'Government public health legislation was the main reason for progress in combating lung cancer in the period c1950 to the present.' How far do you agree? [16+4 marks]

Stimulus: **Doll and Hill's research (1950)** | **Technological treatments (radiotherapy, surgery, immunotherapy)** *(You must also use information of your own)*

ACCESSIBLE WRITING FRAMES:

On the one hand, government legislation was decisive because... • For example, ... • This was significant because... • On the other hand, high-tech treatments mattered because... • In particular, ... • Overall, the most decisive factor was... because...

KEY VOCABULARY BANK:

Richard Doll • Austin Bradford Hill (1950) • British Doctors Study • 1965 TV ad ban • 2007 public smoking ban • plain packaging (2016) • radiotherapy • immunotherapy

Write your response in the space provided below: *(Response continues on facing page)*

TIMELINE MISSION Turn to Pages 2–3 (Key Topic 4.5). In the drawing box, sketch a 1950s doctor cigarette advert and a 2016 olive-green plain pack. Annotate how state legislation slashed smoking from 65% to under 13%! **[ ] Done**

Continue your response to Question 5/6 below:

• **QUESTION 3: EXPLAIN ONE SIMILARITY [4 MARKS]** **SCORE: [/ 4] • 5 MINS**

Explain one way in which government intervention against lung cancer after 1965 was similar to government action in the 1875 Public Health Act. [4 marks]

Focus on both representing state intervention and compulsory legislation replacing laissez-faire.

Compare 1875 sewers with 2007 smoking bans: state compulsion consistently achieves greater public health impact than laissez-faire.

KEY TOPIC 4 NOTES & EXAMINATION STRATEGY

Notes: Record key causal links, essay plans, and teacher feedback for this unit.

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

ENQUIRY	ASSESSMENT COMPONENT FOCUS	PAGE	SCORE	TEACHER COMMENT & NEXT STEPS
KT4.1	**Q4:** Explain Why Rapid Progress in Causes c1950–Present	p. 5–6	/ 12	
	Q3: Explain One Difference (Investigating Causes: 1880s vs 1990s)	p. 6	/ 4	
KT4.2	**Q5/6:** Evaluative Essay on Diagnostic Scanners vs Lifestyle Campaigns	p. 8–9	/ 20	
	Q3: Explain One Difference (Diagnostic Methods: 19th c vs Late 20th c)	p. 9	/ 4	
KT4.3	**Q5/6:** Evaluative Essay on the NHS as a Medical Turning Point (1948)	p. 11–12	/ 20	
	Q3: Explain One Difference (Healthcare Access: 1930s vs Post-1948)	p. 12	/ 4	
KT4.4	**Q4:** Explain Why Penicillin was Mass-Produced (1928–1945)	p. 14–15	/ 12	
	Q3: Explain One Similarity (Fleming 1928 vs Jenner 1796)	p. 15	/ 4	
KT4.5	**Q5/6:** Evaluative Essay on Government Action & Lung Cancer	p. 17–18	/ 20	
	Q3: Explain One Similarity (Anti-Smoking Laws vs 1875 Public Health Act)	p. 18	/ 4	
TOTAL	**Key Topic 4 Cumulative Examination Assessment Portfolio (10 Tasks)**	p. 4–18	/ 104	**Overall Grade:** _____ **Teacher Target:** _____

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES

Interactive Textbook • 10 Questions Per Enquiry

KT4.1
DNA & Genetics
LESSON HUB
PRACTICE QUIZ
Score: [/ 10]

KT4.2
Lifestyle & Scans
LESSON HUB
PRACTICE QUIZ
Score: [/ 10]

KT4.3
Magic Bullets & NHS
LESSON HUB
PRACTICE QUIZ
Score: [/ 10]

KT4.4
Penicillin Study
LESSON HUB
PRACTICE QUIZ
Score: [/ 10]

KT4.5
Lung Cancer Study
LESSON HUB
PRACTICE QUIZ
Score: [/ 10]

From DNA to the NHS, state action and laboratory science revolutionized healthcare; revision guarantees your GCSE success.